The Unilateral Shift Obstructs the Nontrivial Frame-Orbit Decomposition Conjecture

A scoped counterexample to the final conjecture of arXiv:2301.06183

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Status. Candidate counterexample, likely valid, pending expert review. The printed statement is tautological if zero summands are allowed; its natural nontrivial reading, requiring at least two nonzero summands, is false.

1 Source conjecture

For a separable Hilbert space $\mathcal{H}$, the source paper [1] defines

$$E(\mathcal{H}) = \left\{ T \in B(\mathcal{H}) : \{T^k\varphi\}_{k=0}^{\infty} \text{ is a frame for } \mathcal{H} \text{ for some } \varphi \in \mathcal{H} \right\}.$$

On page 11, its final conjecture states that, for every $T \in E(\mathcal{H})$, there is a Hilbert direct-sum decomposition

$$\mathcal{H} = \bigoplus_{n\in\mathbb{N}} \mathcal{H}_n$$

such that (i) $T\mathcal{H}_n \subseteq \mathcal{H}_n$, and (ii) for every n there is $\varphi_n \in \mathcal{H}_n$ for which $\{T^k\varphi_n\}_{k=0}^{\infty}$ is a frame for $\mathcal{H}_n$.

We close this work by raising the following conjecture:

Conjecture. Let $T \in E(\mathcal{H})$. The Hilbert space $\mathcal{H}$ can be decomposed into a direct sum of Hilbert spaces

$$\mathcal{H} = \bigoplus_{n\in\mathbb{N}} \mathcal{H}_n \ ,$$

where

(i) $T\mathcal{H}_n \subset \mathcal{H}_n$.
(ii) For any $n \in \mathbb{N}$, there exists $\phi_n \in \mathcal{H}_n$ such that, $\{T^k\phi_n\}_{k=0}^{\infty}$ is a frame for $\mathcal{H}_n$.

Figure 1: The final conjecture on page 11 of the source paper.

2 Idea of the counterexample

The unilateral shift is already the canonical example of an operator with a frame orbit: the orbit of its first basis vector is the entire standard orthonormal basis. At the same time, the multiplicity-one shift cannot split into two nonzero invariant blocks. Indeed, in an orthogonal decomposition into invariant summands, the complement of each summand is invariant as well, so its orthogonal projection commutes with the shift. But the only orthogonal projections commuting with the multiplicity-one unilateral shift are 0 and the identity.

This proves that the only possible direct-sum presentation has one nonzero summand, namely the whole space. Thus the conjecture has a sharp convention dichotomy: it is automatic if zero summands are admitted and false if the summands are required to give a genuine decomposition.

3 The literal convention

Proposition 1 (Vacuity of the literal statement)**.** *Suppose zero Hilbert spaces are allowed as direct summands and the zero orbit is regarded as a frame for the zero space. Then the printed conjecture is true immediately for every $T \in E(\mathcal{H})$.*

Proof. By the definition of $E(\mathcal{H})$ there exists $\varphi \in \mathcal{H}$ such that $\{T^k\varphi\}_{k\geq 0}$ is a frame for $\mathcal{H}$. Set

$$\mathcal{H}_1 = \mathcal{H}, \qquad \varphi_1 = \varphi, \qquad \mathcal{H}_n = \{0\}, \quad \varphi_n = 0 \quad (n \geq 2).$$

Then $\mathcal{H} = \bigoplus_{n\in\mathbb{N}} \mathcal{H}_n$, every summand is invariant, and the frame inequalities on a zero space are vacuous. □

If the convention is instead that every displayed summand must be nonzero, the next theorem supplies a counterexample.

4 The unilateral-shift counterexample

Let $\mathcal{H} = \ell^2(\mathbb{N}_0)$ with standard orthonormal basis $(e_j)_{j\geq 0}$, and let S be the unilateral shift

$$Se_j = e_{j+1} \qquad (j \geq 0).$$

Theorem 2 (Failure of every nontrivial decomposition)**.** *The operator S belongs to $E(\mathcal{H})$, but $\mathcal{H}$ cannot be written as an orthogonal direct sum of two or more nonzero closed S-invariant subspaces. Consequently, the source conjecture is false under the natural interpretation that its countable family of summands is nonzero.*

Proof. First,

$$\{S^k e_0\}_{k=0}^{\infty} = \{e_k\}_{k=0}^{\infty}$$

is an orthonormal basis for $\mathcal{H}$, hence a Parseval frame. Therefore $S \in E(\mathcal{H})$.

Suppose that

$$\mathcal{H} = \bigoplus_{n\in I} \mathcal{H}_n$$

is an orthogonal direct sum of closed S-invariant subspaces. Fix n and let P_n be the orthogonal projection onto $\mathcal{H}_n$. The orthogonal complement

$$\mathcal{H}_n^{\perp} = \bigoplus_{m\neq n} \mathcal{H}_m$$

is also S-invariant: finite partial sums are carried into the same complement, and boundedness of S passes this to the closure. Thus S is block diagonal relative to $\mathcal{H}_n \oplus \mathcal{H}_n^{\perp}$, so

$$P_n S = S P_n. \tag{1}$$

Because $P_n = P_n^*$, taking adjoints in (1) also gives $S^*P_n = P_nS^*$. Now

$$\ker S^* = \operatorname{span}\{e_0\},$$

so $P_n e_0 = ce_0$ for some scalar c. Since P_n is a projection, $c^2 = c$, hence $c \in \{0, 1\}$. Finally, for every $k \geq 0$,

$$P_n e_k = P_n S^k e_0 = S^k P_n e_0 = ce_k.$$

Therefore $P_n = cI$. Each $\mathcal{H}_n$ is consequently either $\{0\}$ or $\mathcal{H}$, and there cannot be two nonzero summands. In particular, no countably infinite nonzero decomposition satisfying condition (i) exists; condition (ii) cannot repair this obstruction. □

5 Even nonorthogonal topological sums fail

The use of "direct sum of Hilbert spaces" normally denotes an orthogonal Hilbert direct sum, which Theorem 2 handles. The same counterexample survives a broader bounded topological-direct-sum reading.

Proposition 3. *The commutant of the multiplicity-one unilateral shift contains no nontrivial idempotent. Hence $\ell^2(\mathbb{N}_0)$ cannot be a bounded topological direct sum of two nonzero closed S-invariant subspaces either.*

Proof. Identify $\ell^2(\mathbb{N}_0)$ unitarily with the Hardy space $H^2(\mathbb{D})$, so that S becomes multiplication M_z by the coordinate function. If a bounded operator A commutes with M_z, let $\psi = A1$. For every polynomial q,

$$Aq = Aq(M_z)1 = q(M_z)A1 = q\psi.$$

Boundedness and density show that $A = M_\psi$ for a bounded analytic multiplier $\psi \in H^\infty(\mathbb{D})$. If $A^2 = A$, then $\psi^2 = \psi$. The analytic function ψ takes values only in $\{0, 1\}$, so connectedness of $\mathbb{D}$ forces it to be constant. Thus $A = 0$ or $A = I$.

If $\mathcal{H} = M \dotplus N$ were a bounded topological direct sum of two closed S-invariant subspaces, the coordinate projection onto M along N would be a bounded idempotent commuting with S, contradicting the preceding paragraph unless one summand were zero. □

6 Verification and limitations

The counterexample is scoped to the only nontrivial reading of the printed conjecture: at least two nonzero summands. It does not claim that the literal zero-summand convention is false; Proposition 1 shows that version is automatic.

The script `code/verify_shift_obstruction.py` checks the analogous matrix statements for truncated shifts in every dimension from 2 through 40. It verifies that the first-vector orbit has identity Gram matrix and that the joint commutant of the shift and its adjoint is one-dimensional. These 39 finite checks are designed to catch orientation and commutator mistakes; the infinite-dimensional proof above is exact and does not depend on them.

The most important human-review question is semantic rather than analytic: whether the source authors intended all $\mathcal{H}_n$ to be nonzero. Under that intention, Theorem 2 is a complete disproof. Without it, Proposition 1 is a complete literal answer.

7 Bounded novelty check

On 27 August 2026, the run registry, solution, attempt, and proof-gap indexes were searched for arXiv:2301.06183, its title, unilateral-shift frame orbits, invariant-summand decompositions, and reducing-subspace variants. Bounded exact-phrase, arXiv, title, and citation searches found the source paper, the related arXiv papers 2212.01921 and 2505.19303, and a 2025 article citing the source (doi:10.1007/s11785-025-01839-8). No exact answer to the final conjecture surfaced; the available metadata and abstract of the citing paper concern other characterizations. Novelty is plausible, not certified.

References

[1] J. Cheshmavar, A. Dallaki, and J. Baradaran, *On abstract results of operator representation of frames in Hilbert spaces*, J. Pseudo-Differ. Oper. Appl. 14 (2023), article 30, arXiv:2301.06183v1, doi:10.1007/s11868-023-00524-8.